

Supplementary Information

Dispersed awards, persistent ties: evidence from Türkiye’s public IT procurement

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S1 Data construction

S1.1 Source and IT scope

Award records were collected from EKAP (ekapv2.kik.gov.tr) by searching domain keywords. Search matches substrings of tender titles, which admits unrelated tenders. The keyword *erp* retrieved 618 contracts, of which 546 are not IT (it matches, among others, the place name Yaverpaşa and *tuz serpici*, a salt spreader). The phrase *teknik destek* retrieved 1,186 contracts, of which 798 are hospital or university staffing and cleaning contracts. An earlier count-based cleaning had removed 394 obvious false positives from 13,418 records, leaving 13,024.

We re-audited this set by value. The original audit classified 150 contracts ranked by the then-recorded numeric amounts by hand (108 IT, 1 call-centre service, 19 mixed, 22 not IT). These amounts were not yet uniformly converted across currencies; the claim concerns that historical reviewed set. For the remainder, ordered title rules assign each contract to one of four scope classes: IT, IT-enabled call-centre services, ambiguous, and not IT. Works contracts are ambiguous only if they install an IT or operational-technology system, such as city security cameras, SCADA, fibre, or a data centre, and are otherwise not IT. Staffing contracts for data entry are ambiguous; bundled cleaning or catering contracts without an IT system are not IT. Random samples from every rule were inspected, and seven rule outcomes for top contracts were overridden by hand. Table S1 reports the result. The main sample consists of the IT and call-centre classes.

EKAP’s award list has one row per tender identifier, so tenders with several lots appear with a single winner. We found 116 main-sample titles that suggest multiple lots and no framework-agreement call-offs (the word *çerçeve* appears only in the sense of “within the framework of”).

Table S1: Scope classes after the value-based audit (13,024 contracts that passed the earlier count-based cleaning).

Scope class	Contracts	Nominal value (bn TRY)	Share of value (%)
IT	9,981	25.68	56.4
IT-enabled call-centre services	10	7.62	16.7
Ambiguous (excluded from main sample)	1,406	4.13	9.1
Not IT (removed)	1,627	8.12	17.8
Total	13,024	45.54	100.0

Search provenance. The preserved listing checkpoint contains 20,171 distinct tender identifiers and includes all 13,418 winner records, which in turn contain all 13,024 early-cleaned records. First-match query attribution agrees exactly with the released `kaynak_keyword` field. The current collector contains 48 ordered keywords, 50 records per page, a 10,000-result cap per keyword, a 2009 start date, and a runtime-dependent end date. Identifiers are retained only on first encounter across queries; keyword counts therefore do not measure all matching queries. These source-code settings are not a recovered execution manifest: exact extraction dates, query completion, cap hits, and executed end dates are not recorded. The checkpoint and later winner chain do not establish exhaustive collection. The complete keyword list, preserved source settings, source hashes and annual retrieved/retained counts are in `docs/sampling_provenance.md` and `results/revision_e_sampling/annual_sampling_counts.csv`; safe collector excerpts are in `src/collection/ekap_v3_query_provenance.py`. These aggregate files describe the preserved artifacts and are not national coverage estimates.

S1.2 Firms, buyers, and categories

Firms. Names were canonicalized by upper-casing with Turkish locale, removing punctuation and whitespace, and stripping legal-form suffixes from the end of the name, subject to a minimum remaining length. Joint ventures were kept separate from their members. The merge groups of the roughly 300 largest firms were reviewed by hand. Seventy-two brand groups were confirmed, including firms that changed legal form (for example from limited to joint-stock company) or name while keeping the same brand in non-overlapping years. Three uncertain brand pairs were left unmerged. The process reduces 4,788 recorded names to 4,389 firms. By a name-pattern heuristic, 536 of these firms (302 in the main sample) are natural persons. In the released data, every winner that is a natural person or lacks an explicit legal-entity form (690 firms) is replaced by a stable pseudonym.

Buyers. Thirty-six generic buyer labels pool units from two or more provinces, among them the provincial health directorates and hospital unions of the Ministry of Health. We split them by the province recorded for each tender (1,001 of the 13,024 contracts; 34 labels and 732 contracts in

the main sample), which turns 3,453 labels into 3,663 buyer nodes. Six national bodies with offices in two provinces were not split. The recorded labels are used as a sensitivity check throughout.

Buyer sectors. Buyers were assigned to 11 sectors by ordered keyword rules on the buyer name. Relative to an earlier coding, 1,423 contracts changed sector, mainly because upper-casing without Turkish locale had prevented mixed-case names from matching, and because several rules matched unintended substrings.

Product categories. Contracts were assigned to 14 product categories by ordered rules on the tender title, most specific first; in the main sample, 97% receive a specific market. For an agreement audit of the rules, a random sample of 300 main-sample titles (seed 7) was labelled blind to the rules by two separately run AI coders based on different language models, each seeing only the titles and a codebook; an earlier AI check of the same titles is reported for comparison. The two separately run coders agree on 88.7% of titles ($\kappa = 0.87$), and each agrees with the earlier check on 90–92% ($\kappa = 0.89$ – 0.91). The rules agree with the separately run AI coders on 73–75% of titles ($\kappa = 0.70$ – 0.72), and with their joint label on 79.3% of the 266 titles where the two coders agree. Against that joint label, health information systems has precision 1.00 and recall 0.99, GIS 1.00 and 0.94, and software licences 0.80 and 0.97. Among the weakest are custom software (0.48 and 0.56), maintenance (0.60 and 0.41), cybersecurity (recall 0.50), ERP (precision 0.69), and the residual “other IT” class. The labels are AI-produced and have no human gold-standard reference. The 79.3% consensus-only comparison excludes 34 of 300 cases; rule agreement across all 300 is 74.7% with coder A and 73.0% with coder B. Model agreement is not classification accuracy. The replication material includes full class denominators, pairwise confusion matrices, blind sheets for 300 titles and 60 title pairs, and a prospective human-validation protocol. It requires two people’s independent coding, freezing their original labels before reconciliation, and reporting uncertainty and missing/unclear outcomes. No completed human labels are asserted.

Dates, procedures, values. Dates are tender dates joined by tender identifier (coverage 100% within the extract, October 2010–March 2026). The first and last years are partial; this coverage measure does not establish national procurement coverage. Procedures are open, restricted, and negotiated Article 21(a)–(f); the extract contains no direct procurement under Article 22. In the main sample, 67 explicit foreign-currency amounts are converted to nominal TRY using the midpoint of TCMB buying and selling rates on the tender date, or the preceding available business day. Currency is unspecified in 845 records and is assumed to be TRY with an explicit flag. This is not independently verified currency, and the tender date is not necessarily the settlement date. Nominal TRY values are then deflated to 2025 TRY using the annual-average consumer price index (2026 uses the January–August mean of the linked monthly series, although observed 2026 awards end in March). Producer-price deflation is a sensitivity check; count-based statistics are unaffected by either conversion or deflation.

S1.3 Descriptive statistics by buyer sector

Table S2 describes the main sample by buyer sector.

S1.4 Article 21 of Law No. 4734

Translations are ours. The chapeau reads: “The negotiated procedure may be used in the following cases.”

- (a) No tender was submitted in an open or restricted procedure.
- (b) Before 25 May 2018: the tender must be held urgently because of sudden and unexpected events such as natural disasters, epidemics, or a danger of loss of life or property, or of events the contracting authority could not foresee. Law No. 7144 (Art. 11, in force 25 May 2018) inserted: “or in cases that have special features in terms of construction technique, or in which the contracting authority determines that the work must be carried out urgently to ensure structural safety or the safety of life and property”.

Table S2: Buyer sectors (main sample). Value in bn 2025 TRY; procedure shares in % of contracts.

Buyer sector	Contracts	Firms	Buyers	Value	21(b)	21(f)	Open
Health	2,950	554	1,098	30.93	28.5	9.4	61.8
Municipal / local	2,688	1,084	667	20.09	2.2	44.1	53.4
Other public	1,295	646	351	11.03	2.5	41.1	54.7
Education	1,054	519	330	4.68	5.1	31.8	62.6
Infrastructure / transport	644	337	123	7.75	2.8	39.8	57.0
Agriculture / environment	395	231	98	4.30	2.3	31.4	63.5
Social / finance	303	156	42	5.31	3.6	39.3	55.4
Provincial administration	223	125	74	0.99	1.3	25.1	62.8
Security / interior	202	142	64	1.52	3.5	26.2	66.8
Defence	191	96	29	1.89	1.0	3.7	95.3
Justice	46	32	14	1.25	26.1	28.3	43.5
All	9,991	2,814	2,890	89.74	10.5	29.6	59.0

- (c) The tender must be held urgently because of special defence- or security-related circumstances.
- (d) The procurement requires research and development and is not subject to serial production.
- (e) The technical and financial characteristics cannot be specified with the necessary precision because the goods, services, or works are original and complex.
- (f) Purchases of goods, materials, or services whose estimated cost is below a monetary limit, updated every 1 February (for example 157,923 TRY in 2014 and 3,406,508 TRY in 2026).

In cases (b), (c), and (f), no notice needs to be published; at least three firms are invited to submit qualification documents and price offers. In our data all 2,957 main-sample 21(f) contracts are goods or services, and from February 2014 on, 99.96% are below the limit in force.

S1.5 Currency-status sensitivity

The currency-corrected top-150 ranking replaces one record with tender 2023/670417 (eight de-icing-fluid vehicles), already excluded as non-IT. An additional AI-assisted source check matched its identifier and eight-vehicle scope to the official DHMİ tender notice (page 1; tender date on page 3): official DHMİ notice, IKN 2023/670417. This supports exclusion as non-IT, but does not verify the award value and is not independent human annotation. The historical 150-record review must not be described as a completed human review of the corrected ranking. The replication currency-screen file records this change.

The 845 main-sample amounts with unspecified currency account for 5.27% of real value under the TRY assumption. Excluding them changes pooled value HHI from 147.24 to 158.51 and health-information-system value HHI from 831.07 to 872.61. These checks describe sensitivity to the flagged records, not verification of their currency. The original procedure-logit specification did not converge in the explicit-currency-only sample and is not interpreted. A matched-control linear-probability comparison yields log-value coefficients of -0.0038 and -0.0050 , and Article 21(b)-post interaction coefficients of 0.1189 and 0.1384, respectively, in the full and explicit-currency samples ($n = 4,913$ and 4,786). This auxiliary comparison changes both sample and functional form relative to the main logit and is only descriptive.

S2 Concentration

Table S3 reports supplier concentration by product category. Bootstrap intervals are bias-shifted percentile intervals (1,000 resamples of contracts within market). “No max” removes the market’s largest contract; “winsor.” caps contract values at the 99th percentile of the main sample. Table S4

reports medians of HHIs computed within calendar years for market-years with at least 30 contracts. The bias-corrected count index is the unbiased estimator $10,000 \sum_i n_i(n_i - 1)/[n(n - 1)]$, where n_i is firm i 's number of contracts and n is the market-year total.

Table S3: Supplier concentration by product category (main sample; HHI on 0–10,000).

Product category	Contracts	Firms	HHI count [95% CI]	HHI value [95% CI]	Value, no max	Value, winsor.	CR4 value (%)
Health information systems	2,189	205	431 [405, 458]	831 [692, 985]	868	812	53
ERP / management software	1,515	599	84 [71, 97]	131 [74, 218]	119	96	16
Custom software / web / mobile	1,117	734	25 [22, 29]	262 [41, 737]	175	110	27
Software licences	1,047	375	103 [89, 118]	274 [189, 436]	275	269	25
Network / data-centre infrastructure	748	390	72 [60, 87]	280 [151, 523]	233	226	28
Computers / peripherals	696	467	40 [34, 47]	169 [69, 357]	145	155	18
Maintenance / support services	626	297	82 [69, 98]	299 [194, 473]	300	295	24
GIS / city information	553	203	353 [273, 444]	408 [228, 842]	385	347	33
Cybersecurity	390	194	193 [143, 263]	476 [193, 999]	441	341	37
Physical security / surveillance	385	222	84 [69, 104]	212 [123, 373]	208	210	20
Other IT	277	215	79 [58, 111]	757 [0, 2,278]	594	294	46
Education technology	205	128	225 [147, 346]	1,062 [196, 2,936]	752	539	54
Smart city / traffic	169	106	171 [125, 234]	2,448 [483, 5,712]	813	581	68
Call centre / help desk	74	46	690 [363, 1,160]	8,893 [3,517, 10,000]	1,639	1,366	98
All categories pooled	9,991	2,814	39 [36, 41]	147 [40, 501]	91	87	18

Table S4: Median within-year supplier HHI by product category.

Product category	Years	Count HHI	Count HHI, bias-corr.	Value HHI	Value HHI, no max
Health information systems	16	671	597	1,091	1,081
GIS / city information	12	650	369	1,503	1,206
Physical security / surveillance	4	427	108	1,098	926
Cybersecurity	7	416	156	1,666	1,263
Maintenance / support services	14	343	91	1,721	1,238
Software licences	15	325	187	998	868
Network / data-centre infrastructure	15	274	68	1,038	937
Computers / peripherals	15	265	44	903	762
ERP / management software	15	226	119	688	521
Custom software / web / mobile	15	176	37	740	601
All categories pooled	17	69	55	256	209

Decomposition. The pooled index satisfies $H_{agg} = \sum_m w_m^2 H_m + C$, where w_m is market m 's share and $C = \sum_i \sum_{m \neq m'} s_{im} s_{im'} w_m w_{m'}$ collects the terms of firms active in several markets. By count, $H_{agg} = 38.6$, $\sum_m w_m^2 H_m = 26.7$ and $C = 11.9$; $\sum_m w_m^2 = 0.117$ and the demand-weighted mean of market HHIs is 179. By real value the corresponding figures are 147.2, 126.7 and 20.5, with a weighted mean of 1,269 (529 without the largest contract). A quarter of firms are active in more than one market; they hold 69% of contracts.

S3 Incumbency and null models

S3.1 Subgroups and sensitivities

Definitions are given in the main text. Tables S5–S9 report incumbency by subgroup against the same permutation draws restricted to the subgroup. The nulls are N1 (market $\times$ year), N2 (adding buyer sector), and N3 (adding buyer province instead). All subgroup tests under N1 have $p \leq 0.001$, except for four very small groups: restricted procedure ($p = 0.24$, $n = 8$), Article 21(a) ($p = 0.04$, $n = 7$), and Article 21(d) and the year 2010 (one eligible contract each). Table S10 reports sensitivities; Figure S1 shows incumbency by year.

Further robustness checks give the following results. Excluding the 116 possible multi-lot tenders gives an observed incumbency of 0.455 (N1/N2/N3 nulls 0.062/0.109/0.209). Excluding natural-person winners gives 0.464 (0.066/0.116/0.214). The excess amounts to 1,919, 1,683, and 1,203 incumbent contracts under N1, N2, and N3.

Table S5: Incumbency by product category.

Product category	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Health information systems	1,252	0.716	0.151 [0.136, 0.167]	0.157	0.407
Maintenance / support services	375	0.536	0.029 [0.013, 0.045]	0.139	0.122
Physical security / surveillance	182	0.418	0.030 [0.011, 0.055]	0.097	0.317
Other IT	112	0.393	0.019 [0.000, 0.045]	0.132	0.233
ERP / management software	745	0.392	0.027 [0.016, 0.039]	0.077	0.111
Network / data-centre infrastructure	328	0.384	0.023 [0.009, 0.040]	0.106	0.121
Smart city / traffic	58	0.397	0.041 [0.000, 0.103]	0.104	0.314
Cybersecurity	161	0.398	0.052 [0.019, 0.087]	0.132	0.186
Computers / peripherals	281	0.327	0.012 [0.000, 0.025]	0.047	0.164
Software licences	602	0.334	0.039 [0.025, 0.055]	0.105	0.139
Education technology	80	0.350	0.068 [0.025, 0.125]	0.201	0.168
Call centre / help desk	19	0.316	0.062 [0.000, 0.158]	0.222	0.178
GIS / city information	230	0.326	0.088 [0.057, 0.122]	0.117	0.146
Custom software / web / mobile	488	0.201	0.006 [0.000, 0.014]	0.029	0.056

Table S6: Incumbency by buyer sector.

Buyer sector	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Health	1,502	0.667	0.129 [0.116, 0.142]	0.143	0.372
Other public	583	0.395	0.030 [0.017, 0.045]	0.075	0.120
Education	439	0.364	0.028 [0.014, 0.043]	0.118	0.164
Municipal/Local	1,271	0.362	0.031 [0.023, 0.041]	0.057	0.143
Provincial admin	115	0.374	0.046 [0.009, 0.087]	0.108	0.309
Social/Finance	196	0.357	0.041 [0.015, 0.066]	0.185	0.086
Agriculture/Environment	210	0.348	0.042 [0.019, 0.067]	0.110	0.100
Infrastructure/Transport	342	0.322	0.029 [0.015, 0.047]	0.097	0.116
Defence	131	0.298	0.039 [0.008, 0.076]	0.219	0.070
Security/Interior	100	0.280	0.025 [0.000, 0.060]	0.145	0.103
Justice	24	0.292	0.038 [0.000, 0.125]	0.292	0.091

S3.2 Nulls that preserve firm geography

A firm's prior modal award province is the buyer province where it won most contracts before the tender; this is not its registered location; firms without earlier wins form their own category. Null N4 permutes winners within market $\times$ year $\times$ winner's prior award province: the null mean is 0.173 (ratio 2.61, excess 1,370 contracts). Null N4b permutes within market $\times$ year $\times$ buyer province $\times$ winner's prior award province: the null mean is 0.264 (ratio 1.71, excess 923 contracts). Under N4b, 38.4% of eligible contracts are in cells with a single distinct observed winner, which makes it highly restrictive. Recurring purchases have an incumbency of 0.720 against an N4b null of 0.391, other purchases 0.305 against 0.195 (all $p \leq 0.001$). A version with province proxies computed from all wins, including the contract itself, gives null means of 0.185 (N4) and 0.242 (N4b).

S3.3 Named-product licences and maintenance

Titles that indicate licence renewal, maintenance, support, or updates of a named product (for example hospital information systems, Oracle, Microsoft) account for 10.5% of repeat-eligible contracts. Their incumbency (0.486) is close to that of other contracts (0.448). Excluding them, recurring purchases have an incumbency of 0.723 against an N3 null of 0.296 (2.44 times), and other purchases 0.306 against 0.163 (1.88 times). Excluding all titles with any licence, maintenance, support, or update wording (44% of eligible contracts) leaves ratios of 2.07 and 1.77.

Table S7: Incumbency by buyer type (keyword classes of buyer names).

Buyer type	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
MoH hospital / dental centre	617	0.831	0.177 [0.154, 0.203]	0.181	0.386
University hospital / medical faculty	110	0.627	0.099 [0.045, 0.155]	0.120	0.363
MoH provincial directorate / hospital union	695	0.563	0.098 [0.079, 0.118]	0.113	0.391
University (non-health unit)	401	0.362	0.025 [0.012, 0.040]	0.113	0.164
Municipality / municipal company	1,271	0.362	0.031 [0.023, 0.041]	0.057	0.143
Other	296	0.351	0.031 [0.014, 0.051]	0.086	0.183
Other health body (central/agency)	82	0.378	0.058 [0.012, 0.110]	0.139	0.130
Central government / agency	1,441	0.353	0.035 [0.027, 0.045]	0.124	0.107

Table S8: Incumbency by procedure.

Procedure	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Article 21(b)	618	0.723	0.143 [0.120, 0.165]	0.159	0.376
Article 21(c)	34	0.412	0.020 [0.000, 0.088]	0.069	0.255
Open	3,041	0.441	0.057 [0.051, 0.064]	0.112	0.203
Article 21(f)	1,204	0.346	0.033 [0.023, 0.043]	0.080	0.131
Article 21(a)	7	0.286	0.049 [0.000, 0.286]	0.173	0.227
Restricted	8	0.125	0.033 [0.000, 0.125]	0.059	0.035
Article 21(d)	1	0.000	0.001 [0.000, 0.000]	0.000	0.000
Article 21(e)	0	–	– [–, –]	–	–

S3.4 Recurring

Titles are normalized by Turkish-aware lower-casing, removal of diacritics, digits, month names, and about 110 generic procurement words, and stemming to five characters; similarity is the Jaccard index of the stemmed word sets. A contract is flagged as a recurring purchase if the buyer has a contract in the same market 183–548 days earlier with similarity of at least 0.5. Sixty best-match pairs, drawn at random within four similarity bands (15 each), were coded blind by the same AI coders. Of the 42 pairs above the threshold, 35–40 were judged the same recurring service (depending on the coder), but so were 6–10 of the 18 pairs below it. Agreement between the rule and the coders is moderate ($\kappa = 0.40$ – 0.45), and between the two separately run coders $\kappa = 0.63$. These are model judgments on a stratified pair sample, not a human estimate of population precision or recall. Rewording can produce false negatives and generic titles false positives; the direction of resulting bias in the persistence contrast is unknown. The flag denotes a recurring purchase, not a legal contract extension, and the unflagged group is not necessarily composed of newly arising needs. Table S11 reports results by procedure and market and alternative definitions.

S3.5 Buyer dependence

Figure S2 shows the distribution of main-supplier shares. With per-buyer permutation p -values and the Benjamini–Hochberg procedure at nominal 5% across the 540 buyers, 273 buyers (50.6%) exceed their N1 null on the main-supplier share and 357 (66.1%) on the supplier HHI; under N2 the figures are 212 (39.3%) and 289 (53.5%). The joint dependence conditions required for a formal false-discovery-rate guarantee are not established for these shared permutation draws. With 1,000 permutations, no buyer can pass a Bonferroni correction. The share of buyers above their 95th null percentile (uncorrected) rises with buyer size (50% for 5–9 contracts, 63% for 10–19, 80% for 20 or more). The mean excess, however, falls with size (0.24, 0.23, 0.13), so the rising share reflects the tighter null distributions of large buyers, that is, statistical power.

Table S9: Incumbency by year.

Year	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
2026	46	0.587	0.079 [0.022, 0.152]	0.151	0.301
2025	353	0.450	0.067 [0.045, 0.093]	0.112	0.222
2024	350	0.557	0.078 [0.054, 0.106]	0.144	0.263
2023	413	0.496	0.098 [0.075, 0.121]	0.131	0.265
2022	463	0.501	0.080 [0.058, 0.102]	0.132	0.260
2021	462	0.450	0.068 [0.050, 0.089]	0.106	0.222
2020	410	0.483	0.076 [0.054, 0.100]	0.131	0.224
2019	307	0.518	0.072 [0.046, 0.101]	0.122	0.262
2018	345	0.441	0.050 [0.029, 0.072]	0.095	0.147
2017	363	0.355	0.046 [0.028, 0.066]	0.082	0.148
2016	320	0.409	0.043 [0.025, 0.066]	0.087	0.174
2015	310	0.410	0.041 [0.023, 0.065]	0.098	0.167
2014	295	0.400	0.035 [0.017, 0.054]	0.091	0.165
2013	234	0.410	0.039 [0.017, 0.064]	0.084	0.148
2012	165	0.352	0.024 [0.006, 0.048]	0.060	0.137
2011	76	0.368	0.022 [0.000, 0.053]	0.118	0.136
2010	1	0.000	0.029 [0.000, 1.000]	0.033	0.000

Table S10: Sensitivity of the incumbency result (1,000 permutations each; all $p \leq 0.001$).

Variant	Observed	Null mean [95% interval]	Ratio
Main (N1: market $\times$ year)	0.452	0.062 [0.056, 0.067]	7.3
N1 with 3-year windows	0.452	0.060 [0.055, 0.066]	7.5
N2: market $\times$ year $\times$ sector	0.452	0.110 [0.103, 0.116]	4.1
Buyer = recorded label	0.462	0.106 [0.100, 0.112]	4.4
Firm = recorded name	0.394	0.047 [0.042, 0.052]	8.4
Excluding call-centre contracts	0.452	0.062 [0.056, 0.067]	7.3
Including ambiguous scope	0.435	0.056 [0.051, 0.061]	7.8
Cells = buyer sector $\times$ year	0.452	0.047 [0.042, 0.052]	9.7
Incumbency at buyer level (any market)	0.389	0.053 [0.049, 0.058]	7.3
Excluding Article 21(f) contracts	0.495	0.079 [0.071, 0.087]	6.3
N3: market $\times$ year $\times$ province	0.452	0.207 [0.201, 0.213]	2.2

S3.6 Category and observed-history stress tests

We specified seven sensitivity variants before running their permutations and used 1,000 permutations with seed 20260927. All fix contracts, buyer identities, dates, and supplier win totals in category–year–buyer–province cells, except the category-free variant, which removes category from both incumbency and permutation restrictions. Merging ERP, custom software, maintenance, and other IT is a taxonomic stress test, not an assertion that these form one economic market. Later evaluation windows retain the complete 2010–2026 observed history in both observed and permuted series. Health-only and no-other-IT variants restrict the sample before histories are computed.

The replication output saves all null draws, cell-mobility shares, and full class-level AI audit denominators. These are correlated sensitivity checks inside the captured sample. They cannot estimate national keyword-search recall, restore pre-2010 history, or eliminate time-varying capture. The sample begins in October 2010 and ends in March 2026.

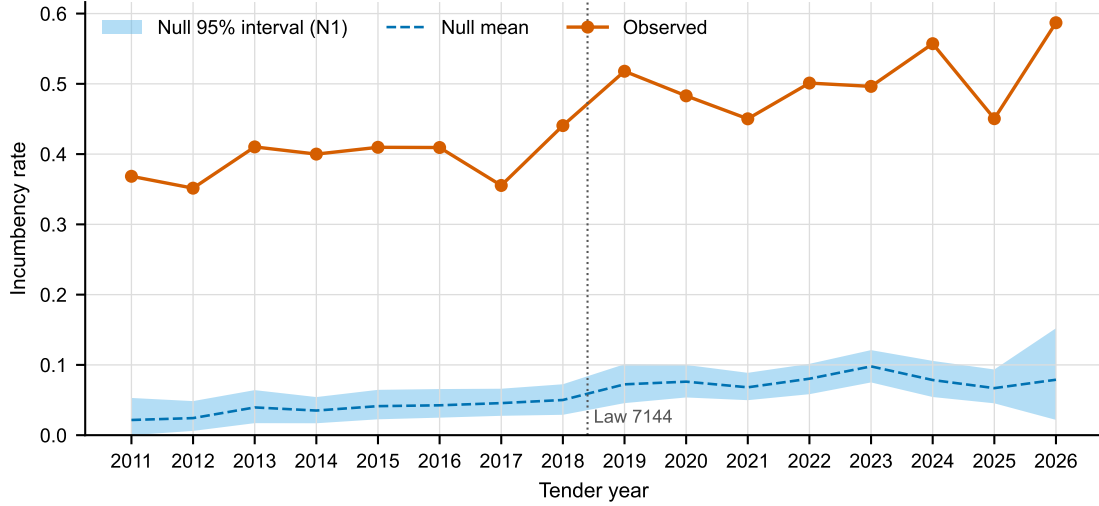


Figure S1: Incumbency rate by year, observed and under null N1 (mean and 95% interval).

S4 Who wins: conditional associations and diagnostics

Choice sets. For each repeat-eligible contract, the alternatives are the firms that won at least one contract in the same product category in the 24 months strictly before the tender date; covariates use only information before the tender. In 38.7% of the 4,913 repeat-eligible contracts the winner is not in this set (an entrant to the market or returning after more than 24 months): 65% of non-incumbent wins, 20% of recurring purchases, 49% of other purchases, and 52% of contracts under other procedures (mainly 21(f)). The main model uses the remaining 3,013 contracts (909 buyers). At least one of the buyer’s incumbents is in the choice set in 87.5% of these contracts, and the most recent supplier in 85.8%. An earlier version of the choice sets included firms whose only wins came after the tender. That produced an artificial “no prior wins” effect and is not used.

Estimates. Table S13 reports the incumbent odds ratio across specifications. Adding each excluded observed winner back to its choice set gives 21.5 (18.5–25.0), a sensitivity estimate with outcome-dependent choice-set augmentation, not a bound. An indicator for such winners cannot be estimated because it is one only for chosen alternatives. With a pool that includes firms active in any market, the incumbent odds ratio is 31.9 and firms new to the market have an odds ratio of 0.23.

Future-tie diagnostic. We add an indicator for a firm with no strictly earlier award from the buyer in the category but an award on a strictly later date. Its odds ratio is 54.4 (42.2–70.1), alongside 63.7 for past suppliers; restricting outcomes to 2023 or earlier gives 52.4. These are outcome-conditioned associations. Current wins can affect later ties, so the diagnostic is not an exogenous negative control. The dynamic-history simulation redraws 3,013 winners chronologically while recomputing incumbency; the main version retains other observed covariates and awards outside the modeled sample. Its future-tie odds ratio is 19.8 (16.1–24.5). The mismatch with observed data shows that this particular benchmark does not reproduce all moments; it does not estimate a heterogeneity share. Among flagged recurring purchases, past and future odds ratios are 258 and 128; among other purchases they are 30 and 40.

Firm-activity and first-future-award checks. Adding future wins in the category from other buyers (odds ratio 1.40 per log unit) reduces the future-tie odds ratio from 54.4 to 42.0 (32.3–54.5); adding past wins from other buyers as well gives 41.5. These controls use outcomes too and do not identify a dyadic causal component. In the corrected lag check, the first strictly future award, not any later award, must occur more than 365 days after the focal tender. The corresponding odds ratio is 46.64 (34.33–63.36), or 33.58 (24.47–46.06) with firm-activity controls. A separate indicator captures first future awards within the following year. Same-day awards do not establish past or future status.

Table S11: Incumbency of recurring purchases and other purchases: observed [N1 null; N3 null].

	Recurring	Other
All	0.720 [0.092; 0.288] ($n=1,743$)	0.305 [0.045; 0.163] ($n=3,170$)
Article 21(b)	0.900 [0.198; 0.427] (291)	0.566 [0.093; 0.331] (327)
Open	0.669 [0.077; 0.267] (1,128)	0.307 [0.045; 0.165] (1,913)
Other (mainly 21(f))	0.735 [0.049; 0.237] (324)	0.210 [0.026; 0.099] (930)
of which no similar earlier title	–	0.188 [0.034; 0.104] (2,117)
Health information systems	0.869 [0.209; 0.452] (519)	0.607 [0.110; 0.375] (733)
ERP / management software	0.821 [0.034; 0.227] (201)	0.233 [0.024; 0.069] (544)
Software licences	0.593 [0.048; 0.212] (241)	0.161 [0.034; 0.091] (361)
Custom software	0.544 [0.010; 0.152] (79)	0.134 [0.006; 0.038] (409)
Maintenance / support	0.776 [0.036; 0.138] (192)	0.284 [0.021; 0.107] (183)
Network / data centre	0.670 [0.035; 0.184] (112)	0.236 [0.016; 0.090] (216)
Computers / peripherals	0.436 [0.014; 0.194] (110)	0.257 [0.010; 0.146] (171)
GIS / city information	0.828 [0.135; 0.274] (58)	0.157 [0.071; 0.104] (172)
Threshold 0.40	0.709 [–; 0.286]	0.296 [–; 0.159]
Threshold 0.67	0.744 [–; 0.302]	0.329 [–; 0.168]
Window 0–18 months	0.695 [–; 0.293]	0.268 [–; 0.143]
Window 0–24 months	0.684 [–; 0.290]	0.250 [–; 0.135]

Table S12: Classification and observed-history sensitivity checks. All upper-tail Monte Carlo $p = 1/1,001$.

Variant	Eligible	Observed	Null mean	Ratio
Baseline	4,913	0.4523	0.2073	2.182
Merged software services	5,397	0.4334	0.1848	2.345
Category-free buyer history	7,087	0.3893	0.1033	3.768
Health information systems	1,252	0.7157	0.4069	1.759
Evaluate from 2013; retain history	4,671	0.4573	0.2109	2.168
Evaluate from 2015; retain history	4,142	0.4640	0.2177	2.132
Exclude other IT	4,801	0.4537	0.2066	2.195

Equal observation horizons. A further diagnostic uses equal 730-day past and future windows and only focal dates for which both windows lie fully inside the observed extract. Same-day awards are excluded. Past status means a buyer–firm award in the preceding 730 days, rather than ever before; future-only status requires an award in the following 730 days and no award in the past window, so older ties may still exist. The original pre-tender 24-month alternative sets and full-history controls are retained. For 2,508 contracts, past and future odds ratios are 52.63 (43.70–63.38) and 53.74 (39.76–72.62). Their ratio is 0.979 (0.767–1.251); by recurrence status it is 1.675 (1.108–2.533; $n = 1,177$) and 0.484 (0.346–0.678; $n = 1,331$). Qualitative differences therefore persist with bounded observation opportunities, but the future indicator still conditions on outcomes and has no causal interpretation.

S4.1 Simulation benchmarks and initialization

Dynamic-history benchmark. Winners of the 3,013 modeled contracts are redrawn chronologically from the fitted conditional logit, with incumbency recomputed from simulated history. Awards outside the modeled sample retain their observed winners. With other covariates held at observed values, the future-tie odds ratio is 19.8 (16.1–24.5); recomputing history-based covariates gives 24.2 (20.0–28.9); drawing coefficients from their estimated distribution gives 19.8 (15.3–25.8). These versions retain observed choice-set restrictions and portions of observed history, so they do not generate an entire procurement economy under pure state dependence. Table S14 reports the examined coefficient grid. The calibration reproducing the modeled incumbent share (0.689) has coefficient 3.93

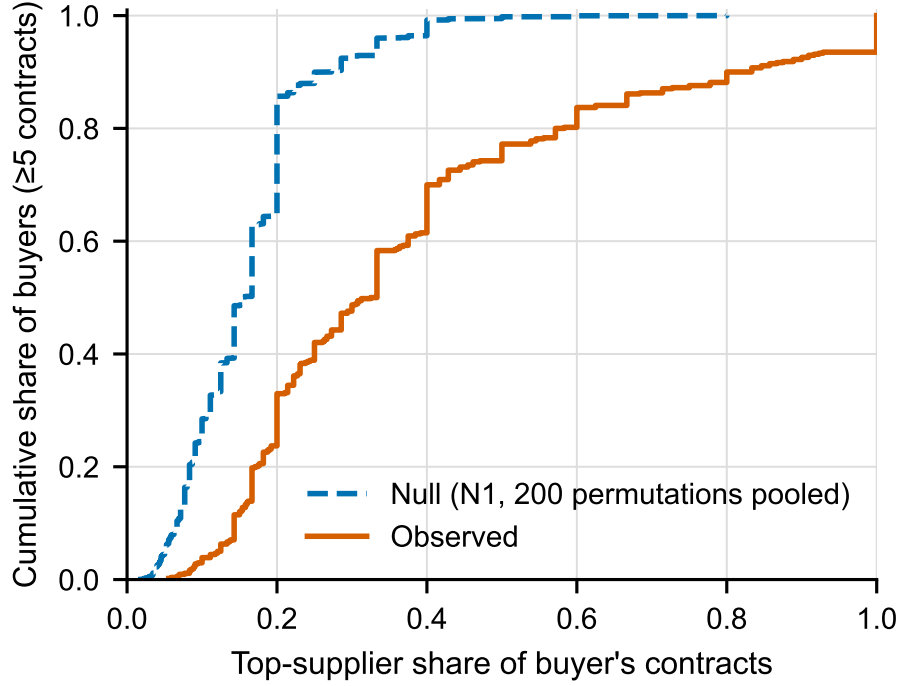


Figure S2: Distribution of the main supplier’s share of each buyer’s contracts (540 buyers with at least five contracts), observed and under null N1.

and future-tie odds ratio 21.9 (17.9–26.5). No examined grid value reproduces both moments; this is not a general impossibility result.

Outcome-conditioned random-intercept benchmark. The direct incumbency coefficient is zero and persistent buyer–firm intercept dispersion is calibrated to the observed incumbent share ($\sigma = 2.92$ overall, 4.44 for flagged recurring purchases, 2.47 for others). Crucially, initialization is an outcome-conditioned heuristic using observed history. It is not a posterior sample from an estimated likelihood and does not solve the initial-conditions problem. Consequently, the benchmark cannot be read as independent evidence that heterogeneity alone explains the observed economy. Its full-sample incumbent share is 0.457 versus 0.452 observed. Table S15 and Figure S3 compare algorithmic moments, conditional on these initialization and calibration choices.

No dynamic-history replication reaches the observed past/future ratio in any of the three samples. With only 100 replications, a plus-one empirical tail calculation cannot resolve probabilities below 1/101. Standardized distances from simulation means are not causal tests or precisely estimated tail probabilities. The random-intercept algorithm also generates recency ratios above one despite its zero direct incumbency coefficient; because it is initialized using observed outcomes, this observation cannot isolate the effect of heterogeneity. The estimated last/earlier ratio after controlling for prior dyad wins is 1.69 (1.38–2.07).

Mixed-coefficient sensitivity. Additional grids set the direct incumbency coefficient to 0, 1, 2, or 3 and recalibrate intercept dispersion (50 replications). In the original outcome-conditioned initialization, coefficients 2–3 (odds ratios 7–20) place some recurring-purchase moments nearer the observations. This range is neither an estimated switching-cost parameter nor a confidence interval. Other-purchase and pooled moments are not jointly reproduced by the same settings. Such comparisons illustrate specification sensitivity; they do not establish that switching costs operate only in recurring purchases or that unflagged purchases exhibit no state dependence.

Initialization sensitivity with common recomputed histories. We repeat pooled simulations with zero explicit same-category incumbency coefficient, fixed fitted coefficients on other covariates, observed alternative pools, and 6,978 observed fixed awards. All historical covariates are now recomputed in every scenario. The 3,013 modeled awards are simulated jointly across purchase

Table S13: Conditional logit: incumbent odds ratio [buyer-clustered 95% CI].

Specification	Contracts	Incumbent OR
24-month pool, all active firms (main)	3,013	42.3 [36.4, 49.2]
Recurring	1,388	140.7 [107.0, 185.1]
Other	1,625	21.0 [17.6, 24.9]
Article 21(b)	529	95.8 [69.4, 132.3]
Open	1,877	33.7 [28.1, 40.5]
Other procedures (mainly 21(f))	607	43.7 [32.3, 59.1]
24-month pool, 30 random alternatives	3,013	41.8 [35.8, 48.8]
24-month pool, 10 random alternatives	3,013	44.5 [36.3, 54.5]
12-month pool, all active firms	2,456	39.8 [33.8, 46.9]
Observed winner added (30 alternatives)	4,913	21.5 [18.5, 25.0]
Recurring / other purchases		48.1 / 13.0

Table S14: Dynamic-history benchmark: moments by incumbency coefficient (other coefficients at fitted values; 100 replications).

β_{inc} (odds ratio)	Share of incumbent winners	Future-supplier OR [95%]	Past/future ratio	Last/earlier ratio
2.745 (15.6)	0.510	10.0 [7.8, 12.3]	2.02	1.01
3.245 (25.7)	0.590	14.3 [11.4, 17.9]	2.36	0.99
3.745 (42.3), fitted	0.665	19.8 [16.1, 24.5]	2.88	1.01
3.93 (51), calibrated	0.691	21.9 [17.9, 26.5]	3.19	1.00
4.245 (69.8)	0.730	26.6 [20.2, 33.7]	3.62	1.00
4.745 (115.0)	0.782	35.1 [27.8, 45.1]	4.56	1.01
Observed	0.689	54.4 [42.2, 70.1]	1.17	1.75

types. For each buyer–firm pair, a normal intercept has standard deviation σ and mean $\sigma^2 I$, where I is an initialization indicator. The all-fixed rule sets $I = 1$ for any pair winning a fixed award anywhere in the observed period, including after modeled choices. The pre-initial rule uses only fixed awards strictly before that buyer’s first modeled award, across categories. The unshifted rule sets $I = 0$ for all pairs. The all-fixed rule shifts 6,436 pairs; the pre-initial rule shifts 1,886. Thus 4,550 pairs receive the legacy shift without a qualifying pre-initial fixed win. This is an audit of outcome conditioning, not a posterior model of initial matches.

Dispersion is calibrated to the modeled incumbent share of 0.689 using 20 common-random-number replications per trial, followed by 100 evaluation replications per scenario. The shifted rules match the target within the specified 0.01 tolerance at $\sigma = 2.90625$ and 3.40625 . The no-shift rule does not attain it on the tested grid through $\sigma = 10$ (largest calibration mean 0.350); we report its evaluation at that ceiling explicitly as unmatched. This is not a global non-attainment theorem or evidence against heterogeneity generally: leaving initial effects unconditioned also imposes a restrictive assumption. Table S16 shows that initialization changes both calibration and lead–lag contrasts. The all-fixed label refers only to the legacy shift mask; unlike the earlier random-intercept results, all covariates here are recomputed, so comparisons should be made within this new exercise. These conditional replication ranges omit parameter, calibration, and specification uncertainty. Neither shifted rule solves initial conditions or identifies a switching-cost effect.

S5 Procedure and timing

S5.1 Contract-level model

The 21(b) $\times$ post-2018 odds ratio is 2.42 (1.43–4.09) with the buyer-history controls and two-way clustering, and 2.42 (1.57–3.73) with buyer clustering. Without the prior-contract and prior-supplier controls it is 2.90 (1.64–5.15); without any buyer-history controls it is 2.93 (1.65–5.19). The pre-2018 marginal effect of 21(b) is between -3.5 and -3.8 percentage points and insignificant in all

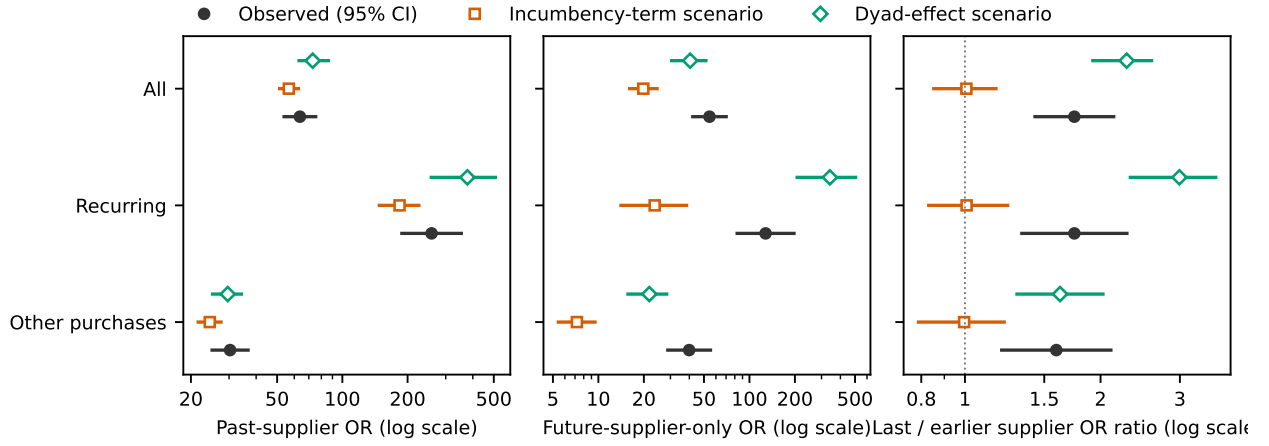


Figure S3: Observed choice-model moments (95% confidence intervals) and specified simulation benchmarks (mean and central 95% range across 100 replications). The dynamic-history benchmark uses an incumbency coefficient; the random-intercept benchmark sets it to zero but uses outcome-conditioned initialization. These are algorithmic comparisons, not identified pure mechanisms. Plot labels abbreviate the two specifications. Log scales.

Table S15: Observed moments (95% CI) and algorithmic benchmarks (mean [2.5–97.5%], 100 replications). The random-intercept benchmark uses outcome-conditioned initialization.

Sample	Moment	Observed	Dynamic history	Random intercept
All	Future-supplier OR	54.4 (42.2–70.1)	19.8 [16.1, 24.5]	40.5 [30.6, 51.5]
	Past/future ratio	1.17 (0.95–1.44)	2.88 [2.48, 3.37]	1.82 [1.45, 2.22]
	Last/earlier ratio	1.75 (1.43–2.14)	1.01 [0.85, 1.17]	2.29 [1.92, 2.60]
Recurring	Future-supplier OR	128 (83–197)	23.6 [14.1, 38.4]	341 [207, 504]
	Past/future ratio	2.02 (1.43–2.85)	8.18 [5.29, 12.24]	1.14 [0.84, 1.51]
	Last/earlier ratio	1.75 (1.34–2.29)	1.01 [0.83, 1.24]	2.99 [2.33, 3.60]
Other	Future-supplier OR	39.9 (28.8–55.2)	7.2 [5.4, 9.5]	21.7 [15.7, 28.3]
	Past/future ratio	0.76 (0.57–1.01)	3.44 [2.62, 4.22]	1.39 [1.04, 1.82]
	Last/earlier ratio	1.60 (1.21–2.11)	1.00 [0.79, 1.22]	1.63 [1.30, 2.02]

specifications. Further checks give interaction odds ratios between 1.76 and 2.43: an IT-only sample, broad scope, recorded buyer labels, recorded firm names, a linear trend instead of year fixed effects, and excluding 2020. Alternative break dates give 2.55 (25 August 2017, Decree-Law No. 694; log-likelihood -2801.1 against -2802.5 for 25 May 2018), 2.10 (placebo, May 2016, $p = 0.009$), and 1.57 (placebo, May 2020, $p = 0.08$). The interaction is 2.64 among health buyers and 0.86 ($p = 0.78$) among other buyers. In a model with a triple interaction (two-way clustered), the post-2018 21(b) interaction is 2.44 ($p = 0.003$) for health buyers and 0.89 for others. The difference is not significant (odds ratio 2.74, 95% CI 0.79–9.51, $p = 0.11$; in percentage points +13.9, -7.6 to 35.5), because only 84 non-health 21(b) contracts are repeat-eligible. Under Article 21(f), 34.6% of repeat-eligible contracts go to an incumbent, against 13.1% under N3 and 3.3% under N1.

S5.2 Event study

Figure S4 reports the 21(b)–open gap by year relative to 2013–17. In that period the gap does not differ across years (joint tests $p = 0.19$ for all buyers and 0.20 for health buyers). In 2018 it is 37 percentage points higher for all buyers (24–50) and 43 points higher for health buyers (23–62). From 2019 on it averages 14 points above the reference for all buyers ($p = 0.003$) and 12 points for health buyers ($p = 0.04$). With buyer fixed effects, health buyers show a gap averaging -10 points in 2013–17, positive in every year from 2018, and 24 points (10–38) above the reference from 2019 on.

Table S16: Initialization sensitivity with all historical covariates recomputed. Means [2.5–97.5% replication ranges], 100 draws per scenario; the no-shift row is an unmatched grid-ceiling sensitivity, not an equivalently calibrated model.

Initialization	σ	Incumbent share	Future-tie OR	Past/future ratio
Observed	–	0.689	54.4	1.17
All-fixed-award shift	2.906	0.694	43.0 [32.8, 54.6]	1.60 [1.30, 2.02]
Pre-initial-award shift	3.406	0.692	40.4 [30.8, 55.4]	2.06 [1.58, 2.51]
No shift (unmatched target)	10.000	0.348	104.8 [89.9, 123.1]	0.31 [0.27, 0.35]

Among parametric shapes, a step in August 2017 combined with a declining trend fits best. A step in May 2018 fits worse by 3.4–7.8 AIC points across the three samples, and a pure trend fits about as badly as no change. Only 8 and 10 repeat-eligible 21(b) contracts fall in the windows 25 August–31 December 2017 and 1 January–24 May 2018. The years 2010–12, with little buyer history and 24 21(b) contracts, show a high gap as well. Against that base, no later year is significantly higher.

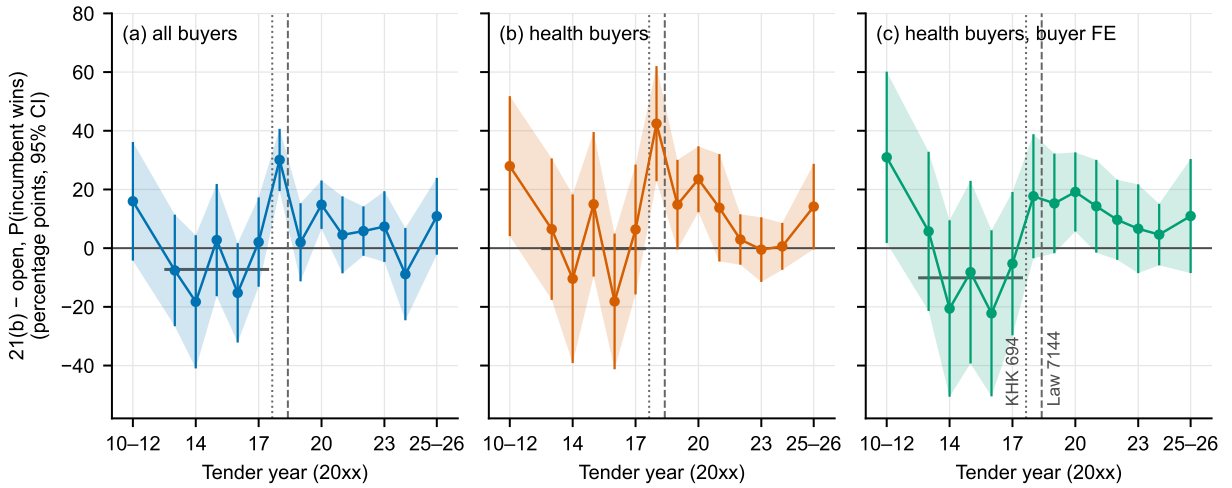


Figure S4: Difference in the probability that an incumbent wins between Article 21(b) and open tenders, by year. Linear probability models with product-market, year, and buyer-sector fixed effects (buyer fixed effects in panel c), log value, and buyer-history controls; two-way clustered 95% intervals. Horizontal bars: 2013–17 reference mean. Dotted and dashed lines: Decree-Law No. 694 (25 August 2017) and Law 7144 (25 May 2018).

S5.3 Dyad continuation

For each new buyer–firm pair in a product category, we ask whether the pair signs another contract in that market (Tables S17 and S18). Of 7,761 new buyer–firm pairs within product categories, 4,334 have no later contract by the buyer in that market and cannot continue. The analysis uses the remaining 3,427, of which 34.6% continue. Covariates are measured at the first contract without look-ahead. The number of later opportunities (the buyer’s later contracts in the market) enters as a covariate, because the offset restriction is rejected (coefficient 0.90 in Poisson pseudo-maximum likelihood, $p = 0.02$; 0.88 in a negative binomial (NB2) model, $p = 0.006$). In NB2 the dispersion parameter is estimated ($\alpha = 0.46$, SE 0.12; likelihood-ratio statistic for $\alpha = 0$: 152.5). Firm size and prior distinct buyers are highly collinear ($r = 0.99$; variance inflation factors of 63 and 61), and generalism correlates 0.86 with size, so these variables are not separately identified in the full model. Entered alone, generalism and size are both positively associated with continuation. There is no evidence that specialist firms are more likely to keep a buyer. A first contract under 21(b) is

associated with continuation (odds ratio 1.85).

Table S17: Dyad continuation, logit (full specification; product-market and year fixed effects; two-way clustered 95% CIs).

	OR [95% CI]	<i>p</i>
First contract under 21(b) (vs open)	1.85 [1.37, 2.49]	<0.001
First contract other negotiated (vs open)	1.26 [1.03, 1.53]	0.026
Log real value of first contract	1.02 [0.95, 1.10]	0.576
Log firm prior contracts (other buyers)	1.58 [0.97, 2.58]	0.067
Log firm prior markets (generalism)	1.20 [0.86, 1.66]	0.283
Log firm prior distinct buyers	0.69 [0.41, 1.15]	0.156
Log buyer prior contracts	0.76 [0.69, 0.85]	<0.001
Log later opportunities of buyer in market	1.89 [1.68, 2.13]	<0.001

Table S18: Number of later contracts, NB2 with estimated dispersion (full specification).

	IRR [95% CI]	<i>p</i>
First contract under 21(b) (vs open)	1.61 [1.39, 1.87]	<0.001
First contract other negotiated (vs open)	1.19 [1.02, 1.39]	0.030
Log real value of first contract	1.01 [0.96, 1.06]	0.706
Log firm prior contracts (other buyers)	1.36 [1.07, 1.74]	0.012
Log firm prior markets (generalism)	1.20 [0.98, 1.46]	0.073
Log firm prior distinct buyers	0.76 [0.59, 0.99]	0.045
Log buyer prior contracts	0.77 [0.72, 0.82]	<0.001
Log later opportunities of buyer in market	2.42 [2.23, 2.63]	<0.001

S6 Single bidding

Random sample. Bid counts were retrieved from published result announcements for a random sample of tenders stratified by year (40 per year drawn from all retrieved tenders). Of these, 393 are in the main sample and report bid counts; none is available for 2010–2011. Rates use year weights based on the main-sample population divided by the number of complete bid-count observations in that year (conditioning on obtained main-sample counts within each year), with linearized 95% intervals (Table S19). Incumbency status is undefined when the tender is the buyer’s first in the market. Adjusting for a negotiated-procedure indicator and a linear year trend (unweighted logit, buyer-clustered, $n = 208$ tenders with defined incumbency), the odds ratio of a single bid for incumbent winners is 4.44 (2.39–8.25). In an alternative unweighted specification with three procedure groups (open, Article 21(b), and other) and a linear year trend, the incumbent odds ratio is 3.66 without and 3.62 with a recurrence control for a single bid, and 2.63 and 2.95 for a single valid bid. The single-bid rate has a weak upward trend (odds ratio 1.05 per year; $p = 0.04$ unweighted, $p = 0.11$ weighted). In 308 tenders with a recorded number of firms that obtained the documents, design-weighted averages are 4.0 such firms and 1.9 valid bids. In 45% of these tenders two or more firms obtained the documents but only one valid bid was received.

Software sample. An interrupted enrichment run of the scraper retrieved result details for 3,400 tenders found with the first search keyword, *yazılım* (software). For tenders from 2021 on, these include the number of valid bids. Joined to the main sample, 1,194 tenders have this count, about 94% of software-titled tenders from 2021 on but only 35% of all tenders in that period. The sample is therefore not random. Single valid bids occur in 66.7% of incumbent-won tenders and 49.3% of others. In a logit with procedure, year, and market fixed effects (681 tenders with defined incumbency, buyer-clustered), the incumbent odds ratio is 2.04 (1.40–2.96), and 2.65 when recurrence status is added (recurrence odds ratio 0.53); without market fixed effects the figures are

Table S19: Single-bid rates (design-weighted random sample).

Group	Tenders	Single bid, % [95% CI]	Single valid bid, %
All tenders	393	44.2 [39, 49]	55.8
Open	225	40.8 [34, 48]	55.9
Article 21(b)	50	75.0 [60, 86]	81.6
Article 21(f)	114	40.0 [31, 50]	46.9
Other negotiated	4	0.0	18.5
Winner is incumbent	95	71.2 [61, 80]	76.7
Winner not previous supplier in market	113	32.4 [24, 42]	48.4
Buyer's first contract in market	185	38.3 [31, 46]	50.2
Before 25 May 2018	140	38.6 [31, 47]	49.1
After 25 May 2018	253	48.9 [42, 55]	61.5

1.99 and 2.68. Controlling for recurrence raises the incumbent effect because recurring purchases are much more common among incumbent winners (55% against 13%), and recurring purchases won by newcomers are the most contested tenders (29% single valid bids, against 52% for other purchases won by newcomers).

S7 Estimated costs and discounts

Sample. From the 4,913 repeat-eligible contracts, 75 won by incumbents and 75 by other firms were drawn at random (seed 11). An AI agent opened each tender's detail page on EKAP in the authors' ordinary browser session, at a pace of roughly one tender per 40 seconds with pauses when the site slowed. It recorded the estimated cost, contract value, highest and lowest bids, product codes, and bid counts, without firm names, and its input list did not show incumbency. It did not interact with any verification widget. It collected 144 tenders before the site displayed a human-verification check, at which point collection stopped. One contract was in US dollars; 143 have both values in lira. EKAP contract values match our records within 1% for 99.3% of matched tenders.

Discounts. The discount is $1 - \text{contract value} / \text{estimated cost}$. It averages 14.2% (median 10.4%, range -9.9% to 84.6%); Table S20 reports it by winner type.

Table S20: Discount on the estimated cost by winner type.

Group	<i>n</i>	Mean	Median	Zero discount
Incumbent winner	73	0.121	0.089	0.082
Other winner	70	0.165	0.112	0.043
Recurring, incumbent	46	0.120	0.087	0.043
Recurring, other	9	0.156	0.019	0.000
Other purchase, incumbent	27	0.123	0.101	0.148
Other purchase, other	61	0.166	0.132	0.049

In OLS with robust standard errors, the incumbent coefficient is -0.044 (SE 0.027, $p = 0.11$) without controls, -0.060 (0.038, $p = 0.11$) with recurrence, procedure, log value, and year, and -0.050 (0.041, $p = 0.22$) with product-market fixed effects as well. Bid counts on the detail pages are available mainly for recent tenders (68 of 144); in that small subsample incumbent-won tenders have 1.8 valid bids on average against 1.3 for others, the opposite of the pattern in the larger bid samples, and we do not interpret it.

Product codes. Among the 143 tenders, the first listed OKAS (CPV) code is in division 72 (IT services) for most markets, including most licence and maintenance contracts; divisions 48 (software packages) and 50 (repair and maintenance) appear mainly among the latter. Division-level codes do not discriminate between our product categories, which are finer.

S8 Network structure

Graph. The main-sample graph has 2,814 firms, 2,890 buyers and 7,219 weighted edges, with density 8.9×10^{-4} . The giant component contains 4,446 nodes (78%) and 92% of contracts. Degree distributions (number of distinct partners) are heavy-tailed. Fitted by maximum likelihood with the protocol of Clauset, Shalizi and Newman (2009), firms have $\hat{\alpha} = 2.21$, and a lognormal is not distinguishable from a power law ($R = -1.67$, $p = 0.09$). For buyers ($\hat{\alpha} = 2.06$) the power law is rejected by the bootstrap goodness-of-fit test, and a lognormal fits better ($R = -6.19$, $p \leq 0.001$). Degree assortativity is -0.17 against -0.04 in the null. Bipartite clustering exceeds the null: Robins–Alexander 0.026 against 0.012, Latapy 0.31 against 0.25. Figure S5 shows the core of the network.

Communities. Over 50 Louvain runs with random seeds and node orders, weighted modularity of the giant component ranges from 0.760 to 0.764 (34–43 communities). Barber’s bipartite modularity of the same partitions is practically identical. In 200 bipartite degree-preserving randomizations (curveball), the unweighted modularity of the giant component is 0.694 ± 0.003 against an observed 0.731, and the weighted modularity is 0.756 ± 0.003 against an observed 0.761. Communities align with buyer sectors (normalized mutual information, NMI, 0.25; permuted labels 0.03; purity 0.64 against a largest-label baseline of 0.44), and ten communities consist of at least 91% health buyers. They align only weakly with firms’ main product categories (NMI 0.13; permuted 0.04). In annual full graphs, modularity exceeds its bipartite null only modestly (annual z between 0.4 and 3.9). Within annual giant components it does not exceed the null in 16 of 17 years, so we do not interpret yearly modularity as evidence of persistent segmentation.

Projections and brokerage. We computed exact Poisson-binomial p -values for co-occurrences under the bipartite configuration model (BiCM). With a false-discovery-rate correction over all pairs at 0.05 or 0.01, no link is validated in either projection, because the data are sparse: 94–97% of co-occurring pairs share exactly one neighbour. A liberal variant that corrects only over co-occurring pairs retains 15,473 firm links and 42,005 buyer links. It essentially reproduces the raw projection without hub-mediated links, so we do not treat it as a validated backbone. Disparity-filter backbones of Newman-weighted projections at $\alpha = 0.05$ contain 1,077 firm links (modularity 0.73) and 2,685 buyer links (0.73). They overlap little with the liberal BiCM set (Jaccard 0.04).

Across these projections and the raw projection, firm betweenness correlates with the number of product categories a firm serves, but also with its number of buyers. Controlling for buyers and contracts, the partial Spearman correlation is 0.12–0.15, and adding the number of markets to a size-only regression raises R^2 by 0.003–0.015. Brokerage thus adds little beyond firm size. In the raw buyer projection, the most connected buyer, a provincial health directorate once pooled labels are split, shares at least one supplier with 19% of the projection’s giant component. With recorded pooled labels the figure is 45%, which shows how strongly such statistics depend on buyer coding.

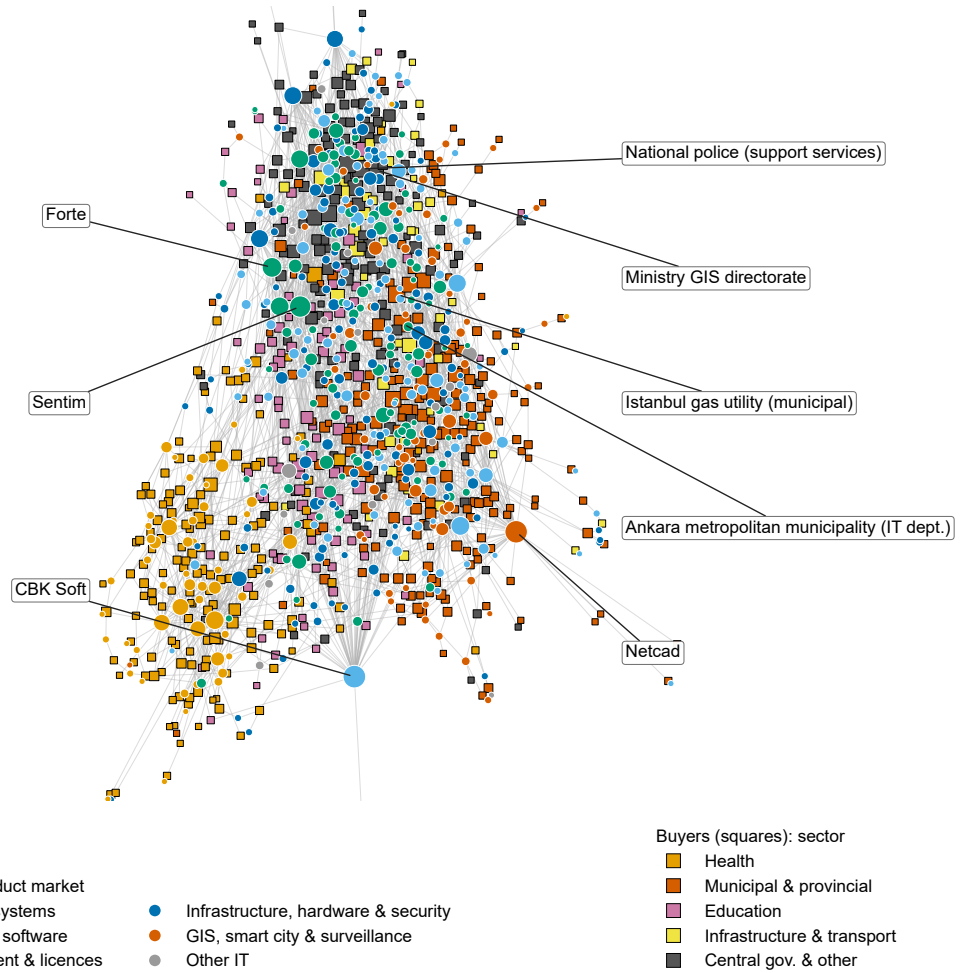


Figure S5: Core of the buyer–supplier network (nodes of degree ≥ 3 , largest connected component; force-directed layout with fixed seed). Circles are firms coloured by their main product category; squares are buyers coloured by sector.